

Laporan Mini Project — Web Server + Database Server Terpisah

Workshop Administrasi Jaringan · PENS — Teknik Informatika dan Komputer Kelompok X = 2 · Tanggal: 2026-06-13 · OS: Debian 13.4 (trixie)

1. Ketentuan & Pemenuhan

Ketentuan	Pemenuhan
VM1 Web Server: NGINX + PHP-FPM + Webmin, IP 10.20.X.10	cli01 10.20.2.10 ✓
VM2 DB Server: MariaDB + Webmin, IP 10.20.X.11	adm01 10.20.2.11 ✓
database TIDAK di localhost	MariaDB bind 10.20.2.11 ✓
✓ web konek ke MariaDB di VM lain	mysqli ke 10.20.2.11 ✓

2. Arsitektur

Host/Klien → HTTP:80 → VM1 cli01 (NGINX+PHP-FPM) → mysqli TCP:3306 → VM2 adm01 (MariaDB)	
10.20.2.10	10.20.2.11
Webmin :10000	Webmin :10000

LAN-B 10.20.2.0/24, gateway/DNS srv01 (10.20.2.1).

3. Database Server (VM2 — adm01)

- MariaDB 11.8 aktif, `bind-address = 10.20.2.11` → listen `10.20.2.11:3306` (BUKAN 127.0.0.1).
- DB `labapp`, tabel `visitors`.
- User remote dibatasi per-host: `webapp'@'10.20.2.10` dengan privilege minimum.
- Webmin aktif di `:10000`.

4. Web Server (VM1 — cli01)

- NGINX + PHP-FPM (php8.4) aktif; vhost `webapp` root `/srv/data/apps/webapp`.
- Aplikasi PHP `index.php` koneksi ke DB remote (sesuai contoh soal): `php $conn = new mysqli("10.20.2.11", "webapp", "WebApp#1206", "labapp");`
- Webmin aktif di `:10000`.

5. Bukti Koneksi DB Remote (bukan localhost)

`mysql -h 10.20.2.11 -u root -p` melaporkan "**10.20.2.11 via TCP/IP**" — koneksi jaringan ke VM lain. Test langsung `mariadb -h 10.20.2.11 ...` dari VM1 → "WEB->DB REMOTE OK". POST form web → INSERT row id=3 tersimpan di DB VM2.

Evidence: `miniproject/evidence/web_to_db_remote.txt`, `webapp_output.txt`, `db_listen.txt`.

6. Cara Akses

- Web app: `http://10.20.2.10/` atau `http://webapp.lab2.local`.
- Webmin VM1: `https://10.20.2.10:10000` · Webmin VM2: `https://10.20.2.11:10000` (login user sistem `fishir`; cert self-signed, terima peringatan browser).

7. Keamanan

- MariaDB bind ke IP LAN-B saja; user DB per-host + privilege minimum.
- PHP prepared statement (anti SQLi) + `htmlspecialchars` (anti XSS).
- Header nginx `nosniff` & `X-Frame-Options DENY`.
- Catatan produksi: simpan kredensial via secret manager/env, aktifkan TLS untuk koneksi MariaDB.

8. Cron Job (penjadwalan otomatis)

Tiga cron job dipasang lewat `/etc/cron.d/` (format dengan field user).

#	VM	Jadwal	Script	Fungsi
1	adm01 (DB)	<code>0 2 * * *</code> (02:00 harian)	<code>/usr/local/sbin/ db_backup.sh</code>	Backup <code>labapp</code> via <code>mysqldump -> /srv/backup/ mariadb/*.sql.gz</code>
2	cli01 (Web)	<code>*/5 * * * *</code> (tiap 5 menit)	<code>/usr/local/sbin/ web_db_healthcheck.sh</code>	Cek koneksi web -> DB remote, log UP/DOWN
3	adm01 (DB)	<code>30 2 * * *</code> (02:30 harian)	<code>/usr/local/sbin/ db_backup_cleanup.sh</code>	Retensi: hapus backup > 7 hari

Bukti berjalan otomatis: log health check terisi pada `00:00:01` oleh cron (`*/5`), bukan eksekusi manual. Backup menghasilkan file `.sql.gz` valid (berisi `CREATE TABLE visitors` + data). Log di `/var/log/db_backup.log` dan `/var/log/web_db_health.log`.

Evidence: `evidence/cron_db_backup.txt`, `evidence/cron_web_healthcheck.txt`. Script & cron file di `scripts/` dan `configs/cron/*.cron`.

9. Bukti Screenshot

[1] Aplikasi web (browser) — menampilkan data dari DB remote

Mini Project - Buku Tamu

Web Server (VM1): 10.20.2.10 (cli01)

Database Server (VM2): 10.20.2.11 (adm01) - MariaDB

Status koneksi DB: TERHUBUNG (host info: 10.20.2.11 via TCP/IP)

Tambah Tamu

Nama: Catatan:

Daftar Tamu (dari DB remote)

ID	Nama	Catatan	Waktu
3	TestPost	insert via web ke DB remote	2026-06-13 23:27:38
2	Kelompok 2	web server konek ke DB remote	2026-06-13 02:36:40
1	Fishir	entri awal mini project	2026-06-13 02:36:40

[2] VM2 Database Server (adm01) — MariaDB :3306 + Webmin + data

```
== VM2 DATABASE SERVER (adm01) ==
IP: 10.20.2.11
mariadb: active
webmin: active
-- listen 3306 (bind ke LAN-B, BUKAN localhost) --
10.20.2.11:3306
-- user remote & data --
User      Host
webapp    10.20.2.10
id        name
1         Fishir
2         Kelompok 2
3         TestPost
fishir@adm01:~$
```

[3] VM1 Web Server (cli01) — NGINX/PHP-FPM/Webmin + test DB remote

```
== VM1 WEB SERVER (cli01) ==
IP: 10.20.2.10
nginx: active
php-fpm: active
webmin: active
-- koneksi PHP web -> DB REMOTE (10.20.2.11) --
Status koneksi DB:</b> TERHUBUNG (host info: 10.20.2.11 via TCP/IP)
-- test mysql langsung ke DB remote --
hasil
WEB->DB REMOTE OK
fishir@cli01:~$ _
```

[4] Cron job DB server (adm01) — backup + retensi + log

```
== CRON JOB - DB SERVER (adm01 10.20.2.11) ==
--- /etc/cron.d/miniproject-db ---
0 2 * * * root /usr/local/sbin/db_backup.sh
00 2 * * * root /usr/local/sbin/db_backup_cleanup.sh
-- backup tersimpan ---
labapp_2026-06-13_235615.sql.gz
-- log backup (db_backup.log) ---
2026-06-13 23:56:15 OK backup -> /srv/backup/mariadb/labapp_2026-06-13_235615.sql.gz (4.0K)
2026-06-14 00:00:26 CLN hapus 0 backup lama (>7 hari)
2026-06-14 00:00:34 CLN hapus 0 backup lama (>7 hari)
fishir@adm01:~$ _
```

[5] Cron job Web server (cli01) — health check + log otomatis

```
== CRON JOB - WEB SERVER (cli01 10.20.2.10) ==
--- /etc/cron.d/miniproject-web ---
*/5 * * * * root /usr/local/sbin/web_db_healthcheck.sh
--- log health check web -> DB remote ---
2026-06-13 23:57:16 UP DB 10.20.2.11 reachable, visitors=3
2026-06-14 00:00:01 UP DB 10.20.2.11 reachable, visitors=3
fishir@cli01:~$ _
```

Catatan: Webmin (kedua VM) terbukti aktif lewat status service & listen :10000 pada screenshot konsol, serta halaman "Login to Webmin" (HTTP 200) saat diakses. Screenshot browser Webmin tidak disertakan karena cert self-signed memblokir render headless; akses manual lewat browser tetap berfungsi.

9. Penyimpangan

- OS Debian 13 (trixie), bukan Debian 12 — seluruh tool wajib (NGINX, PHP-FPM, MariaDB, Webmin) tetap terpasang & berfungsi.
- User admin `fishir` (bukan `student`).